Pediatrics

Growth & Immunization

BY

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Growth and its Disorders

Bone Age Estimation

Developmental stages

- Tanner and Whitehouse described __________ stages of development of ossification centres.

- **'Maturity scoring' assigned as follows:**

    - ________ for carpal bones
    - _______ for radius and ulna
    - _______ for phalanges.

Age specific Radiographs

- **Birth:** Radiograph of _______

- **3-9 months:** Radiograph of _________ is most helpful.

- **1-13 years:** A single film of ______________

- **12- 14 years:** Radiographs of ___________________

Eruption of teeth

Primary dentition (Deciduous Teeth)			
Tooth	**Upper Jaw Eruption(Months)**	**Lower Jaw Eruption (Months)**	**Time of Fall (Years)**
Central incisors			
Lateral incisors			
First molar			
Canine			
Second molar			

Permanent dentition		
Tooth	**Upper Jaw Eruption(Years)**	**Lower Jaw Eruption (Years)**
First molar		
Central incisors		
Lateral incisors		
Canine		
First premolar		
Second premolar		
Second molar		
Third molar		

Approximate anthropometric values by age			
Age	**Weight (kg)**	**Length or height (cm)**	**Head circumference (cm)**
Birth		50	34
6 months		65	43
1 year		75	46
2 years		87	48
3 years		95	49
4 years		100	50

Situations indicating a problem or suggest risk on growth curve:

- A child's growth curve crosses a major _______ line or two major __________ lines.
- A sharp ________ or _________ in the child's growth curve.
- A growth curve that remains _______________

Disorders of Growth

Short Stature:

Definition: It is defined as height ____________ centile or more than _____ standard deviation scores (SDS) below the _________ height for age and gender (<-2SDS) according to the population standard.

Causes:

-
- Cystic fibrosis, asthma
- Malabsorption
-
-
- Pseudohypoparathyroidism
-
-
- Psychosocial dwarfism
- Skeletal dysplasia
- Genetic syndromes

Assessment of body proportion:

- The proportionality is assessed by the upper segment (US): lower segment (LS) ratio.

- **Normally, the US:LS ratio is:**
    - _____ at birth
    - ______ at 3 years
    - ______ by 6 years
    - ______ by 10 years
    - ______ in adults

- **Increase in the US:LS ratio** is seen in ________, _____________, and untreated congenital _______________.

- **Decrease in the US:LS ratio** is seen in ________________________ and vertebral anomalies.

Specific Etiologies:

Familial short stature: The child is short as per definition (height <3rd centile) but is normal according to his genetic potential determined by the parent's height.

Constitutional growth delay:

- These children are born with a ________ length and weight and grow normally during the first year.

- Their growth then ___________ so that the height and weight gradually fall ________ the 3rd centile.

- The onset of puberty and adolescent growth spurt is delayed in these children, but the final height is within normal limits.

Feature	Constitutional growth delay	Familial short stature
Height		
Height velocity		
Family history		
Bone age		
Puberty		
Final height		

Tall Stature:

Definition: Height ____________ standard deviations for age and gender according to the population-specific growth reference is considered as tall stature.

Causes:

- Familial tall stature
- Obesity
- Klinefelter syndrome, Marfan syndrome, homocystinuria
-
-
-

Immunization

Types of Vaccine	
Description	**Example**
Live attenuated	
Killed or inactivated	
Toxoid	
Capsular polysaccharide	
Conjugated	
Recombinant	
Viral vector borne	

Vaccination Route

- Locations for **intramuscular (IM)** administration include **anterolateral aspect of thigh** in infants and **deltoid** in upper arm in older children and adults.

- ___________ administration is **avoided** to reduce risk of **sciatic nerve injury.**

- ____________ is particularly **not given** in **gluteal region** as this has **reduced efficacy.**

- ______ vaccines may be given in **same thigh** but separated by **1 inch.**

- Separate sites are used when administering a vaccine and immunoglobulin.

Principles of Immunization

- Different **live** (oral, parenteral, intranasal) vaccines may be given **simultaneously, or 4 weeks apart**.

- Different types of **inactivated or subunit vaccines** may be administered **simultaneously** or at **any interval** between their doses.

- A minimum interval of ________ between ______ doses of the same vaccine is necessary to ensure adequate immune responses. An exception is the ______ vaccine.

- **Live**-and an **inactivated** vaccine can be administered **simultaneously** or at any interval of time.

- Patients should be observed for **allergic reactions** for __________________ after receiving immunization.

- Immunization is **not contraindicated** in minor illness, prematurity, allergies, malnutrition or during infection and antibiotic therapy.

Mission Indradhanush

The mission derives its name from the ________ diseases prevented through these vaccines (BCG, OPV, pentavalent and TT vaccines) and focuses on **complete immunisation of all children** _______ _______________________ and in ___________ women.

Vaccines covered

-
-
-
-
-
-
-
-
-

Age	National Immunization Program
At Birth	
6 weeks	
10 weeks	
14 weeks	
9 months	
16-18 months	
5-6 yr	
10-12 yr	
9-14 yr	
16 yr	

BCG VACCINE

- **Composition:** Live-attenuated M. bovis Calmette-Guérin (Danish 1331 strain in India).
- **Administration:** ____ mL, __________ at left shoulder, forming a wheal.
- **Immunity:** Primarily cell-mediated; __________ antibodies do not interfere.
- **Storage:** 2-8°C; rapid potency loss after reconstitution.
- **Contraindication:**

- **Preparation and dosage:** Reconstituted with sterile normal saline, each dose (0.1 mL) contains __________ live viable bacilli.

Immunological Effects and Efficacy:

- **Induced Immunity:** Primarily induces ____________ immunity.

- **Efficacy:**

    - **Primary infection:** Low protective efficacy (_____%)

    - **Pulmonary infection:** Variable efficacy (_______%)

    - **Overall forms of tuberculosis:** (______%)

    - **Severe forms of tuberculosis** (miliary tuberculosis, tubercular meningitis): ___________ protection and reduces the risk of mortality.

- Since maternal antibodies ________ interfere with cellular immune responses, BCG is given at birth ensuring compliance and early protection.

- Conventionally, BCG vaccine is administered at insertion of the deltoid on the left shoulder to allow easy identification of its scar.

- Intradermal injection using a 26 G needle raises a 5–7 mm wheal.

- Bacilli multiply to form a papule by one week that enlarges to 4–8 mm, ulcerates by 5–6 weeks and heals by scarring by 6–12 weeks.

- Inadvertent subcutaneous injection causes persistent ulceration and ipsilateral axillary or cervical lymphadenopathy.

- Children with severe immunodeficiency may develop disseminated BCG disease 6–12 months after vaccination.

Poliomyelitis Vaccine

Oral Polio Vaccine (OPV)

- _______________ oral poliovirus vaccine (OPV) developed by __________.
- Contains ____________________ as a **stabilizing agent**.
- OPV vaccine is given as two drops oral at birth or < 2 weeks (zero dose); 3 doses at ________ **and** _____ **weeks (primary); one booster dose at _________ months.**

- Breastfeeding and mild diarrhoea are **not contraindicated** for OPV administration.
- Children with immunodeficiency and pregnant women should ______________ the vaccine.

- Serotype ______ is the most immunogenic of the three serotypes and interferes with the immunogenicity of serotypes 1 and 3.

- VAPP is observed chiefly due to OPV _______ (in recipients) and OPV ________ (in contacts)
- **Seroconversion rates** after 3 doses of OPV are **highest for serotype ____** (90%) and lowest for serotype ____ (70%).

Fractional Inactivated Polio Vaccine (fIPV)

- Intradermal fractional Inactivated Polio Vaccine (fIPV) is used as a dose-sparing strategy in immunization programs, especially in countries with limited vaccine supply.
- The fractional dose (fIPV) is administered intradermally (usually _____) and ___ mL intramuscular or subcutaneous.
- Three intradermal fractionated doses at ___weeks, __ weeks and ______ months.
- The standard intramuscular (I/M) IPV dose is 0.5 mL. So, 0.1 mL (fIPV) is ____________ of the 0.5 mL (I/M) dose.

DPT Vaccine

- Diphtheria vaccine contains diphtheria toxin (DT) inactivated by **formalin** and adsorbed on ______________, the **adjuvant**.
- Quantity of toxoid in vaccine, expressed as **limit of flocculation (Lf)** content is _______ Lf of DT, _____ Lf of tetanus toxoid (TT) and _____ IU of whole cel killed pertussis.
- **Composition:** Diphtheria toxoid (DT), Tetanus toxoid (TT), and either: Whole-cell pertussis (DTwP) or Acellular pertussis (DTaP)

- **Dose & route:** ____ mL; _______________ at anterolateral aspect of mid-thigh.

- **Schedule:** DTwP at 6, 10, 14 weeks; Boosters at 15-18 months and 5 years.

- **Preference:** DTwP is preferred for primary immunization due to better immune response.

- **Contraindication:** anaphylaxis after previous dose, ___________ within 7 days of previous dose.

Acellular Pertussis Vaccine (DTaP)

- Since active pertussis toxin and endotoxin are responsible for the high incidence of adverse events associated with DTwP, various types of purified acellular pertussis vaccines have been developed, termed DTaP.

- The available DTaP vaccines contains atleast ____ pathogenic antigens, commonly **pertussis toxin (PT)** and one or more additional pertussis antigens, like **filamentous hemagglutinin (FHA), pertactin**, **fimbrial protein and a nonfimbrial protein.**

- Each dose of the vaccine contains at least ______ (_______ mg) of PT component and __________ of DT.

- DTaP vaccine is not recommended as part of the National Program in India due to its cost.

- **Contraindication for DTaP are the same as for DTwP**; and the vaccine should not be administered if a previous dose of DTwP or DTaP was associated with immediate anaphylaxis, or the development of encephalopathy within 7 days of vaccination.

- These children should **complete immunization with DT** instead of DTwP or DTaP.

MMR Vaccine

- **Composition:** Live-attenuated measles, mumps, and rubella viruses.
- **Mechanism:** Induces mucosal, systemic humoral, and cellular immunity.
- **Dose, route:** _____ mL; ________________ at right upper arm or anterolateral thigh.
- **Schedule:** Two doses: _______ months and _____ months.
- **Storage:** Lyophilised; 2-8^{0}C use within ______ hours of reconstitution.

Hepatitis B Vaccine

- **Composition:** Contains HBsAg produced by recombinant DNA technology in yeast, adsorbed on aluminium salt.
- **Dose:** ____mL (10 µg); 1 mL for immunosuppressed or high-risk patients.
- **Route:** ________________.
- **Site:** Anterolateral thigh or deltoid; avoid gluteal region.
- **Schedule:**
  - **National:**
  - **Catch-up:**
- **Contraindications:** Anaphylaxis after previous dose.

- **Storage:** 2-8°C; do not freeze.

Mother is known HBsAg negative	Vaccination of the child at 6 weeks.
Mother's status is not known	Vaccinate the newborn within a few hours of birth.
Mother is known HBsAg positive	Vaccinate within few hours of birth , along with hepatitis B immunoglobulin (HBIG) within 24 hr of birth

Rotavirus Vaccine

- Rotavirus is the primary cause of diarrhoea in **infants and toddlers**, accounting for 6-45% of **diarrhoea-related hospitalisations** in Indian children.

- Natural infection does not protect against reinfection or severe disease.

Vaccine Types

Box 10.10: Rotavirus Vaccines

Vaccine name	Rotavac	RotaTeq	Rotarix	Rotasiil
Name by type	Indian neonatal (116E); human bovine monovalent	Human bovine pentavalent (RV5)	Human monovalent (RV1)	Bovine pentavalent; thermostable
Number of strains	1	5	1	5
Schedule[1]				
National program	3 doses at 6, 10 and 14 weeks	Not included	Not included	Not included
IAP 2021	3 doses at 6, 10 and 14 weeks	3 doses at 6, 10 and 14 weeks	2 doses at 6 and 10, or 10 and 14, weeks[2]	3 doses at 6, 10 and 14 weeks
Dose, route	5 drops, oral	2 mL (liquid), oral	1 mL (lyophilized); 1.5 mL (liquid), oral	2.5 mL, oral
Catch up	National program: Up to 1 year of age			

Adverse Reactions

- **Common:** Fever, diarrhoea, vomiting.
- **Rare:**

Contraindications

-
-

Precautions

- Avoid vaccination during

Storage

- **Store at:** 2-8°C.
- **Protect from:** Heat and light.
- **Use within**: ___________ of reconstitution or opening.

Human Papillomavirus (HPV) Vaccine

- Vaccines contain self-assembling virus-like particles (VLP) with recombinant Ll, the major capsid protein.

- VLP based vaccines protect against _______ new infections with the included serotypes.

- The vaccines do not protect against serotypes with which infection has already occurred before vaccination.

- Serotype __________ cause 80-85% of invasive cervical cancers.

- **Non oncogenic serotype:** Serotype __________ cause about 90% of anogenital warts.

- Two vaccines are currently licensed:

- ____________ **(HPV ___)** is a ____________ **vaccine** active against HPV strains 6, 11, 16 and 18 while

- _____________ **(HPV___)** is a __________ vaccine targeting only HPV __________

- _________ prevents serotype-related anogenital warts, vulval and vaginal cancers and vulvar intraepithelial neoplasia.

- ____________ additionally protects against anal, oropharyngeal and head and neck cancers caused by HPV.

- Both vaccines are highly immunogenic and persistent protection **for atleast _________**

- The recommended age for initiation of vaccination is ______ yr, with catch up vaccination permitted up to ______ of age .

- Both vaccines are contraindicated in patients with history of hypersensitivity to any vaccine and should be _________ in pregnancy.

Varicella Vaccine

Vaccination is necessary in following high-risk groups:

-
-
-
-
-

- **Storage:** Freeze dried, lyophilized, 2-8^{0}C, use within ____ __________ of reconstitution.

- When used for post-exposure prophylaxis (within _______ of contact), the protective efficacy of one dose of the vaccine is only 70%.

Rabies

Wound management:

Includes immediate irrigation with running water for _________, thorough cleaning with soap and water and application of povidone-iodine, 70% alcohol or tincture iodine.

Post exposure prophylaxis		
Category	**Type of Exposure**	**PEP Recommended**
I		
II		
III		

Rabies immunoglobulin:

- Provides ________ immunity by neutralizing rabies viruses.

- Its dose is _____ U/kg for human immunoglobulin and ___ U/kg for equine immunoglobulin or equine anti-rabies serum.

- This is infiltrated in and around the wound as soon as possible, not later than _______ of injury.

Vaccination		
Essen or WHO		
Updated Thai red cross		

PEP For Previously Vaccinated Individuals:

- Local wound care
- No RIG needed
- 2 doses on Day ____ and Day ____.

PEP For Unvaccinated Individuals:

- Local wound care
- Three IM doses on days ___________

Yellow fever vaccine

- Live attenuated
- **Dose, route:** _____ mL, ______________
- **Site**: Anterolateral thigh or upper arm
- ________ dose ___________ before travel to endemic areas; revaccinate after 10 years.
- **Storage:** 2-8^{0}C; use within __________ of reconstitution

Adverse events following immunization (AEFI)

Vaccine induced: Event caused or precipitated by an active component of vaccine, e.g.,

- ________________ after measles vaccine
- Vaccine associated paralytic poliomyelitis
- BCG related ___________
- Pertussis _______________

Components of vaccines implicated in mediating allergic reactions	
	Yellow fever, measles, MMR, rabies PCEV, influenza killed injectable and live attenuated vaccines
	Influenza, measles, MMR, rabies, varicella, yellow fever and zoster vaccines
	In the rubber of the vaccine vial stopper or syringe plunger
	• DTaP vaccine
	• Hepatitis B • HPV vaccines

Vaccine specific serious adverse events	
Vaccine	**Adverse event**
Oral polio	
Measles	
Measles, Mumps and/ or Rubella	
Tetanus	
Pertussis	
Rotavirus	
Rubella	

Vaccine Storage and Cold Chain

- Vaccines such as OPV, measles and BCG (especially after reconstitution), are sensitive to heat but can be frozen without harm.

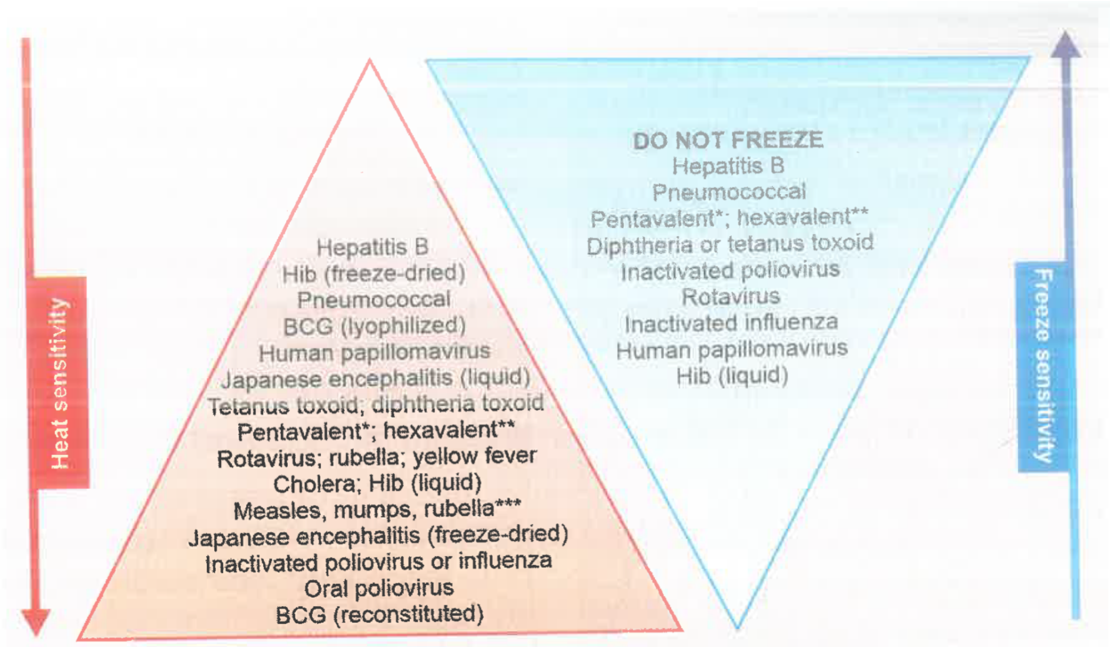
- In contrast, vaccines like hepatitis B, DTP, Pentavalent and tetanus toxoid are _____ sensitive to heat and ________ by freezing.

- In the refrigerator, ________ vials are stored in the freezer compartment (0 to -4°C).

- In the main compartment (4-10°C) _______________ _____ are kept in the top rack (just below the freezer)

- **DTP, DT, TT, hepatitis A and typhoid** are stored in the _________ **racks**

- **Hepatitis B, varicella and diluents** are stored in the __________ **racks.**


Heat sensitivity
Hepatitis B
Hib (freeze-dried)
Pneumococcal
BCG (lyophilized)
Human papillomavirus
Japanese encephalitis (liquid)
Tetanus toxoid; diphtheria toxoid
Pentavalent*; hexavalent**
Rotavirus; rubella; yellow fever
Cholera; Hib (liquid)
Measles, mumps, rubella***
Japanese encephalitis (freeze-dried)
Inactivated poliovirus or influenza
Oral poliovirus
BCG (reconstituted)
DO NOT FREEZE
Hepatitis B
Pneumococcal
Pentavalent*; hexavalent**
Diphtheria or tetanus toxoid
Inactivated poliovirus
Rotavirus
Inactivated influenza
Human papillomavirus
Hib (liquid)
Freeze sensitivity

Immunization limits in unimmunized child according to NIS	
• • • • •	Upto 1 yr of age
• • •	Upto 5 yr of age
	Upto 6 yrs of age
	Upto 7 yrs
	Upto 15 yrs of age